

Plant calendar — API specification design

006_DD-API version 1 supports SCR-006, FN-037–040 and REQ-070–075. WI-010 approved plan revision 1 step 7. Proposed contracts; not implemented.

1. Document control and references

Field	Value
Document ID / version	006_DD-API / 1
System / subsystem	ProductionManagementAI / Master data
Work item / author	WI-010 / Agent
Created / updated	2026-10-02 / 2026-10-02
Review state	Submitted for review; not implemented

Version	Date	Author	Change
1	2026-10-02	Agent	Initial eleven-operation catalog and exact shared field/error contracts

Inputs: [main DD](#), [BD](#), [DB](#), [requirements](#) and [plan](#), all prior designs approved. This document owns HTTP catalogs; later DD-FN owns backend algorithms and DD-SPD owns client processing. Their names are reservations, not broken links. Existing Product/Production lines/order API contracts remain unchanged.

2. Transport, authorization and exact common bounds

Concern	Contract
Base / handler	/api/plant-calendar; PlantCalendarController, all eleven operations
Session / roles	Existing same-origin Identity cookie; Admin or Operator on every read/write, existing ProductionOrderEditor policy; established empty-body 401/403, no HTML redirect
Representation	UTF-8 application/json success; application/problem+json application errors; Cache-Control: no-store on feature responses
Mutation	application/json only, optional UTF-8 charset; JSON object, <=8 KiB including streamed/chunked bodies; depth <=4; no GET/form/query write, batch or bulk endpoint
Strict input	Reject unknown/repeated query keys, unknown/duplicate JSON members, wrong types, missing required properties, invalid Unicode/surrogates and unsupported values; allow only explicitly nullable fields

Concern	Contract
UUID	Nonzero canonical hyphenated UUID text; parse case-insensitively, return lowercase; malformed 400, well-formed unknown reference 404
Date / month	Exact YYYY-MM-DD / YYYY-MM, valid Gregorian date in 0001-01-01..9999-12-31; browser timezone independent; final-month arithmetic must not overflow
Query span	Weekly from/to inclusive, from<=to, at most 366 dates; does not cap persisted future rules or valid single-date capacity/mutations
Pagination	page ASCII 1..10000, default 1; fixed pageSize 20 for rule/history reads, 50 for choices; no client sort identifier
Choice search	q optional, Unicode trim, <=100 code points; literal contains, parameterized; ordering code then UUID using existing DB collation
Version	Canonical decimal JSON string 1..9223372036854775807; no leading zeros, signs, whitespace or JSON number; server-only aggregate revision, opaque in UI
Decimal request	Trim surrounding whitespace, then syntax <code>^(0 [1-9][0-9]*)([0-9]{1,3})?;\$</code> ; >0 and <=24; string only, no sign/exponent/grouping/NaN/infinity/rounding
Decimal response	Canonical invariant non-exponent strings, remove unnecessary trailing zeros; hours 0..24, minutes/unit current valid bounds from WI-009, quantity floor-rounded per unit; never JS float transport
Reason	Required property with nullable string; Unicode trim, blank -> null; <=500 code points after trim, internal content preserved, plain text
Budgets	Feature operation deadline 20s; DB command 15s, statement timeout 15s, lock timeout 5s; cancellation propagated; client timeout 25s; scoped feature configuration, not global rewrites

Date-only future inputs can use the full supported date range. Mutation dates must additionally be >=activation and >=authoritative plant today; future-only withdrawal uses >today. No client counter or If-Match header replaces body version. History/weekly page 2+ requires snapshotVersion equal to the current global revision; page 1 may omit it. Mismatch is 409; paging never silently combines calendar edits. Choice pages describe each read snapshot, not stable selection eligibility forever. Dates before activation are readable as unavailable, never implicitly working. Timezone mismatch returns 503 and blocks only this feature pending operational repair.

3. Operation catalog / registry

All created/updated by Agent on 2026-10-02. Success 200; no redirect/204/201. Operation IDs are new API-PC entries and do not reuse existing API-PL IDs.

ID	Method / relative path	Controller handler / function	Response
API-PC-01	GET /month	Month / FN-037	MonthResult
API-PC-02	GET /day	Day / FN-037/039	DayResult
API-PC-03	GET /weekly	Weekly / FN-038	WeeklyPage
API-PC-04	PUT /weekly/{effectiveFrom}	SaveWeekly / FN-038	MutationResult

ID	Method / relative path	Controller handler / function	Response
API-PC-05	POST /weekly/{effectiveFrom}/withdraw	WithdrawWeekly / FN-038	MutationResult
API-PC-06	PUT /exceptions	SaveException / FN-039	MutationResult
API-PC-07	POST /exceptions/remove	RemoveException / FN-039	MutationResult
API-PC-08	GET /line-choices	LineChoices / FN-037/040	LineChoicePage
API-PC-09	GET /product-choices	ProductChoices / FN-040	ProductChoicePage
API-PC-10	GET /capacity	Capacity / FN-040	CapacityResult
API-PC-11	GET /exception-history	ExceptionHistory / FN-039	ExceptionPage

Each row's full URL is base + relative path. All operations follow: authorize before target lookup; parse strict input/bounds; call the named use case once; map typed outcome to the shared response or Problem Details; record bounded telemetry. This is controller dispatch, not a duplicate of FN processing steps. Reads use one read-only REPEATABLE READ snapshot; writes follow approved DB locks. Select only necessary fields, batch month reads, no per-day DB calls.

4. Shared response field catalogs

A listed property is always present unless expressly optional; nullable values are returned as null, not omitted. Object/array types below have no other fields. Shared schema references in endpoint sections are normative, avoiding repetition.

CalendarContext and scope

Property	Type / requiredness	Source / meaning
plantToday	date string, required	Authoritative operation date basis, e.g. 2026-10-02
timeZone	string, required	Configured IANA timezone, currently Asia/Tokyo
activatedOn	date string or null, required	State activated_on; null if not activated
version	version string or null, required	State revision; null if unactivated
Scope.lineId	UUID or null	Null = plant scope
Scope.line	LineChoice or null	Selected retained line data; plant scope null

LineChoice has id UUID, code string ≤ 50 , name string ≤ 200 , isActive boolean, workingHoursPerDay decimal string. ProductChoice has id UUID, sku string ≤ 50 , name string ≤ 200 , unit from approved seven values and unitRevision canonical bigint string ≥ 0 . Choices contain no edit token or note/credential fields.

DaySummary and source

Property	Type	Mapping / meaning
date	date string	Requested real calendar date
state	Working / Closed / Unavailable	Resolved rule or absent recorded coverage
hours	decimal string or null	Closed "0"; selected line/explicit working hours; line-dependent/unavailable null
hoursBasis	Explicit / LineCurrent / LineDependent / Closed / Unavailable	No invented plant baseline or historical-hours assertion
source	Source object	Winning rule
fallback	Source array, 0..2	Lower applicable rules in priority order; explanation, no partial merge
editable	boolean	Server date/activation/scope eligibility observation, not a permanent grant

Source object: kind LineException/PlantException/Weekly/None; revisionId UUID or null; effectiveFrom date or null; reason string or null. Exception date uses DaySummary.date, effectiveFrom null; weekly has effectiveFrom and null reason; None has all nullable fields null. Source IDs identify immutable snapshots. Before activation state Unavailable, hours null, source None and fallback [].

WeeklySnapshot and ExceptionSnapshot

Object / property	Type	Stored mapping / notes
Both.id	UUID	Immutable revision identity
Both.commitRevision	version string	Recorded aggregate revision
Both.isCurrent	boolean	Latest snapshot for logical target
Both.createdAtUtc	UTC ISO8601 string	Immutable recording instant
Weekly.effectiveFrom	date	Immutable logical start
Weekly.workingDays	weekday string array or null	Mon/Tue/Wed/Thu/Fri/Sat/Sun in weekday order; withdrawn null; all closed []
Weekly.isWithdrawn	boolean	Retained withdrawal marker
Exception.lineId	UUID or null	Plant/line scope
Exception.date	date	Immutable logical date
Exception.isWorking	boolean or null	Removed null
Exception.workingHours	decimal string or null	Explicit hours; working inheritance or closed null

Object / property	Type	Stored mapping / notes
Exception.reason	string or null	Normalized plain text; removed null
Exception.isRemoved	boolean	Retained removal marker

Page metadata: page integer, pageSize integer fixed by operation, totalCount integer ≥ 0 , totalPages integer ≥ 0 (empty 0), items array bounded by pageSize. No truncation of long-standing history: page explicitly. Requested page beyond last yields items [] with actual totals. Weekly/history pages include context and snapshotVersion string or null; unactivated has null and zero records.

MutationResult and target

Properties: context CalendarContext; changed boolean; target WeeklySnapshot or ExceptionSnapshot matching the called operation. Known success only after commit. No-op after authoritative version/permission/target validation returns changed=false, unchanged revision and current target. Changed operation returns appended snapshot and new revision. No capacity/month cache in this response; UI refresh is separate. If refresh fails after known success, retain success notice with a read error; do not reinterpret confirmed commit as unknown.

5. Endpoint contracts

API-PC-01 — GET /month

Request query: month required month string; lineId optional UUID. No other keys. Return MonthResult {context, scope, month, days}, days in date order for exactly the requested month's real 28..31 dates. DaySummary per day; no padded adjacent month cells returned. Missing activation gives unavailable days with 200. Unknown line 404; retired selected line is readable. Common validation/errors.

API-PC-02 — GET /day

Query: date required; lineId optional. Return DayResult {context, scope, day, exception}, day DaySummary; exception current ExceptionSnapshot for exact scope/date or null if never configured, including a removed marker for safe re-add. Provides target id + global version for the editor and verification. Past remains read-only; inherited hours describe current setting, not historical capacity. Unknown line 404. No separate permission to expose another calendar.

API-PC-03 — GET /weekly

Query: from/to required dates, span<=366; view optional Current (default) or History; page optional; snapshotVersion conditional as common rule. Return WeeklyPage {context, snapshotVersion, page, pageSize, totalCount, totalPages, items, applicableBeforeFrom}. Current items are latest snapshots with start inside range, including withdrawal markers; History includes retained revisions for starts in range. Order effectiveFrom DESC, commitRevision DESC, id ASC. applicableBeforeFrom is latest nonwithdrawn current WeeklySnapshot strictly before from, else null; it explains predecessor coverage without pretending the range has no rule. 200 empty/unactivated; no hidden all-future/unbounded history response.

API-PC-04 — PUT /weekly/{effectiveFrom}

Route required date, no query. Required JSON properties: version string; targetRevisionId UUID or null; workingDays array of distinct weekday strings, 0..7 entries. All properties required; [] is valid all closed. New logical date uses null target only if no latest snapshot exists; re-add after withdrawal sends that marker id; correction sends latest id. Server maps days to weekday mask. Today replaces only today's start; older existing start not rewritten. Future correction allowed. Return MutationResult/WeeklySnapshot, including no-op rule. 400 invalid fields, 404 unknown target, 409 stale/date/target mismatch, 503 known busy rollback; uncertainty as section 7. No client isCurrent/commitRevision fields.

API-PC-05 — POST /weekly/{effectiveFrom}/withdraw

Route required date, no query. Body exactly {version, targetRevisionId}, both nonnull strings. Current target must match route, not withdrawn, and start>today. Initial activation protected. Appends withdrawal marker, returns MutationResult. Already withdrawn is 409 CALENDAR_TARGET_CHANGED, not a false successful delete. No bulk withdrawal or physical DELETE.

API-PC-06 — PUT /exceptions

No query. Required body fields: version; targetRevisionId UUID or null; lineId UUID or null; date; isWorking boolean; workingHours decimal string or null; reason nullable string. All properties required. Working null hours means line inheritance; explicit working uses exact positive <=24 hours; closed requires hours null. isRemoved is not accepted in save. Never-configured target uses null id; re-add uses latest removal marker id; edit latest snapshot id. Logical scope/date immutable for edit; cannot use an id from another scope/date. Line create/edit requires active line; plant has no invented line requirement. Return MutationResult/ExceptionSnapshot. Unknown line/target 404, retired/stale/ past/identity conflict 409, field errors 400; no automatic overwrite or conversion.

API-PC-07 — POST /exceptions/remove

No query. Body exactly {version, targetRevisionId, lineId, date}; target id nonnull; lineId nullable. Exact current nonremoved exception, today/future. Retired referenced line allowed only for this existing removal. Appends marker; response MutationResult. Removing absent/already-removed/mismatched target does not

silently succeed: 404 unknown id, 409 target changed. Lower rule visible on subsequent read; full original snapshots remain in history.

API-PC-08 — GET /line-choices

Query: q optional; page optional. Return LineChoicePage {context, page, pageSize, totalCount, totalPages, items}. Active lines only; retained retired scope resolves through month/day instead of appearing as selectable new use. No line creation or edit endpoint added. Unactivated calendar still permits master choices, with context activation/version null; timezone mismatch remains feature unavailable.

API-PC-09 — GET /product-choices

Query: lineId required UUID; q/page optional. Return ProductChoicePage with same page shape. Current active line/product/pair and matching unit/unitRevision only. Unknown line 404; retired line returns 200 empty items. No pair reconfirmation, coefficient edit or unit conversion. Search SKU/name literal; stable SKU,id order.

API-PC-10 — GET /capacity

Query: lineId/productId/date required, no other keys. Past date 400 validation; unknown line/product 404. Return CapacityResult {context, line, product, date, availability, unavailableReason, day, minutesPerUnit, quantity}. line LineChoice; product ProductChoice. availability Available/Unavailable; unavailableReason None/NotActivated/LineRetired/ProductRetired/PairMissing/PairRetired/UnitStale/ CalendarUnavailable; quantity decimal string or null; minutesPerUnit decimal string or null. day DaySummary; no quantity total/history/order allocation.

Unavailable priority: NotActivated, LineRetired, ProductRetired, PairMissing, PairRetired, UnitStale, CalendarUnavailable. Available has reason None and valid minutesPerUnit; quantity=floor(hours*60/minutes) for discrete units, floor at 3 decimals for kg/m. Valid closed pair quantity "0"; invalid pair unavailable even if closed. For Unavailable quantity null; minutesPerUnit null so invalid coefficients are not presented as usable. Consistent current snapshot only.

API-PC-11 — GET /exception-history

Query: date required; lineId optional; page optional; snapshotVersion conditional. Return ExceptionPage {context, scope, date, snapshotVersion, page, pageSize, totalCount, totalPages, items, current}. Items immutable ExceptionSnapshot in commitRevision DESC,id ASC order, including markers; current latest snapshot or null independently of page. Past/retired history readable, no capacity claim. Unknown line 404. Used for observation after unknown commit, not success proof.

6. Errors, message mapping and outcome certainty

Application error follows [RFC 9457](#): type absolute URN, status actual HTTP integer, title safe generic English, code stable catalog code, traceId diagnostic string and optional errors field -> array of catalog codes. No raw input, stack/SQL/detail body. Frontend maps codes to centralized Japanese strings; it never displays arbitrary server HTML. Type is `urn:pmi:problem:plant-calendar:<lowercase-hyphenated-code>`. Write failures additionally carry writeOutcome NotApplied or Unknown; missing/ invalid outcome, generic 5xx, network loss or cancellation is treated as Unknown. No write success can be inferred from an observed equal payload.

Code	HTTP	Meaning / write certainty
VALIDATION	400	Strict field/date/precision error; NotApplied
REQUEST_TOO_LARGE	413	Body >8 KiB; NotApplied
UNSUPPORTED_MEDIA_TYPE	415	JSON required; NotApplied
NOT_FOUND	404	Unknown required identity; NotApplied
CALENDAR_STALE	409	Aggregate/snapshotVersion mismatch; NotApplied
CALENDAR_TARGET_CHANGED	409	Latest id, scope/date or removed/withdrawn target mismatch; NotApplied
CALENDAR_PAST_DATE	409	Mutation date now past; NotApplied
CALENDAR_LINE_RETIRED	409	New/edit line exception unavailable; NotApplied
CALENDAR_PROTECTED_RULE	409	Initial withdrawal or invalid old-start correction; NotApplied
CALENDAR_NOT_ACTIVATED	409	Write attempted before explicit activation; NotApplied
CALENDAR_BUSY	503	Confirmed rollback lock/deadline/deadlock failure; NotApplied, Retry-After: 1
CALENDAR_TIMEZONE_MISMATCH	503	Stored/configured date basis differs; NotApplied
CALENDAR_WRITE_UNKNOWN	503	Commit confirmation lost; Unknown, no Retry-After
UNEXPECTED	500	Safe generic unexpected failure; Unknown for writes

Auth 401/403 retain established empty bodies; middleware/proxy failures may not follow feature Problem Details. Client handles these separately. Read transient errors allow explicit retry. NotApplied allows user correction/manual retry; never automatic mutation retry. An abort is not proof the transaction rolled back. After Unknown, use day/weekly/history reads, keep original input and observed revision, block replay. Verification shows current observation; explicit draft discard/accept current state starts a new baseline, never claims old save success. No idempotency ledger, request completion endpoint or exactly-once promise. DD-FN must distinguish known rollback from uncertain commit reliably.

Suggested Japanese catalog phrases (new, not claimed implemented): 「入力内容を確認してください。」 (Check the input.), 「カレンダーが更新されています。再読み込んでください。」 (Calendar changed; reload.), 「保存結果を確認できません。再送せず現在の状態を確認してください。」 (Save result is unknown; inspect current state without resending.). Exact full catalog/focus sequence belongs to SPD; codes defined here remain stable.

7. Instrumentation, security and consistency

Telemetry registration: ActivitySource/Meter ProductionManagementAI.PlantCalendar. Span names `plant-calendar.<handler>` and child resolve/read/write from DD-FN; ASP.NET Core request span route templates only, no raw query/body. Metric `plant_calendar_operations_total` and `plant_calendar_duration_ms` with bounded operation = eleven registry handler names; outcome = success/validation/conflict/not_found/unavailable/busy/unknown/unauthorized/forbidden/unexpected/cancelled. Count one final outcome per request including parser/auth errors; avoid controller/ use-case double count. Capacity domain Unavailable is HTTP 200 with unavailable outcome. No raw dates/IDs/search/notes/coefficients/tokens/cookies in labels or logs. DD-FN specifies internal spans and measurements; implementation must register both source and meter. No external exporter configuration claimed.

Reuse existing cookie/JSON/same-origin posture; no CORS relaxation or new identity provider. Parameterized SQL and strict bounded parsing; notes render text only. Authorization precedes target-specific responses. Owner activation/DDL never available through API. Snapshot writes honor column grants and line-before-state locks. Calendar reads do not change existing session health or order rules.

Requirement	Contract / later verification
REQ-070	Month/day exact dates, unavailable coverage, scope and bounded snapshots
REQ-071	Weekday transport, start identity, paged history, future withdrawal
REQ-072	Strict exception fields, null inheritance, markers and removal confirmation target
REQ-073	Eligible choices and unit-aware current capacity; unavailable vs valid zero
REQ-074	Roles, limits, errors, versions, unknown certainty and bounded telemetry
REQ-075	No changes to orders, starts, due dates or dashboard date windows

Technical choices submitted for API review: eleven routes, schema/enums, 8 KiB/depth/page/span/time budgets, error outcomes and telemetry names. No open business blocker. SQL/use-case proof and API runtime tests not run. Full DB/FN/SPD consistency gate remains later. Submit this source and EN/JA PDFs and wait for approval before writing DD-FN. Approved designs unchanged.